

David J. Fox

Lean 4 formalization, arithmetic research, and the public program called *Opera Numerorum*.

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Public repositories inventoried: 21

Abstract

This packet documents the public research and software record of David J. Fox, using the live GitHub inventory for DavidFox998, public repository metadata, selected repository READMEs, and public arXiv guidance. The record shows an independent program of Lean 4 / Mathlib formalization spanning Diophantine equations, arithmetic geometry, analytic number theory, spectral questions, mathematical physics, complexity, and software infrastructure.

The principal evidence is reproducible source: Lean developments, build instructions, CI workflows, theorem-dependency discussions, public commit histories, and repository-level mathematical documentation. This packet distinguishes three things that should not be conflated: machine-checked statements in a stated formal environment; research architectures and conditional bridges; and broad mathematical claims requiring ordinary expert review. It is offered as a concise factual record for arXiv sponsorship or editorial consideration, not as a request that a sponsor or editor certify a major open-problem claim.

Author statement for sponsorship and

editorial review

David J. Fox is an independent researcher developing public Lean 4 / Mathlib formalizations in number theory, arithmetic geometry, analysis, and mathematical physics. His work is organized under the name *Opera Numerorum*: a collection of repositories intended to make definitions, dependencies, computations, and proof obligations open to inspection.

The author submits individual manuscripts on their precise mathematical content. Associated code is supplied as a reproducible formal artifact. Where a repository contains a scaffold, hypothesis, imported theorem, or real-number transport boundary, that status is to be stated explicitly. The presence of Lean code, a green build, or the absence of sorry is not offered as a substitute for mathematical exposition or independent peer review.

Evidence at a glance

ITEM	PUBLIC EVIDENCE	APPROPRIATE REPRESENTATION
Identity and public record	Public GitHub profile, public repositories, and linked ORCID.	Independent researcher with an inspectable public technical record.
Formal-methods practice	Lean is the primary language in most mathematical repositories; project documentation commonly includes Lake builds, CI, and axiom/dependency discussion.	Public Lean 4 / Mathlib formalization work with reproducibility-oriented tooling.
Opera Numerorum	Shared naming and cross-links connect the repositories around arithmetic, geometry, spectral analysis, and related topics.	A coordinated research and formalization program; not an independent certification body.

ITEM	PUBLIC EVIDENCE	APPROPRIATE REPRESENTATION
Mathematical status	Repository descriptions include strong conclusions as well as explicit scaffolding and conditional work. The Beal README, for example, explicitly identifies incomplete later layers.	State the exact theorem, hypotheses, imported results, and proof scope for each manuscript.

Reading rule for this packet. “Publicly stated” means that the author’s GitHub metadata or README says it. “Formally checkable” means source can in principle be built and audited in the named environment. Neither phrase means that this packet independently validates a theorem, priority claim, or solution of an open problem.

Public repository inventory

The following table covers all 21 public repositories returned by the GitHub public API on the preparation date. The middle column records the publicly described role. The final column gives language suitable for a sponsor or editor: precise, substantive, and no broader than the evidence summarized here.

Arithmetic, geometry, and RH-related formalization

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>beal-conjecture</u> LEAN	Twenty-one-layer Beal development with import-free core modules, Mathlib wrappers, axiom audits, and a Beal-to-Fermat bridge.	A Lean formalization scaffold for Beal’s Conjecture, with auditable foundational cores and a conditional Beal-to-Fermat corollary. The README explicitly states that later layers remain incomplete scaffolding.

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>brothers-desert-proof</u> LEAN	Route D in the author’s four-route RH architecture; describes a symmetry, “35 Brothers,” and Dirichlet-jitter construction.	A Lean development of a proposed symmetry-based route connected to the author’s RH program; its exact theorem scope should be taken from the source and manuscript.
<u>poincare-spectral</u> LEAN	Formal spectral-gap work for the Poincaré homology sphere (S^3/I^*) , described through a specified finite/tail calculation.	A formal spectral calculation for a specified Poincaré-sphere model, rather than a general solution of an undecidable spectral-gap problem.
<u>lindelof-hypothesis-143</u> LEAN	Lean project centered on $(X_0(143))$, $(S_4 = \{2,3,19,191\})$, a stated GRH implication, and a Lindelöf conclusion.	A Lean formalization project studying a stated $(X_0(143))$ -based route to a Lindelöf-type conclusion; all antecedents must be given in the manuscript.
<u>birch-swinnerton-dyer-143a1</u> LEAN	Formal work on the elliptic curve 143a1, including a claimed analytic/algebraic rank comparison and related Hasse/Bost material.	A formalization of specified BSD-related data for elliptic curve 143a1. Any BSD theorem should be stated curve-specifically with all imported inputs visible.
<u>rh-growth-contradiction</u> LEAN	Route C: a growth-contradiction architecture invoking Littlewood (Ω) -behavior, zero repulsion, (S_4) , and a stated RH conclusion.	A Lean formalization of a proposed analytic route toward RH through growth and zero-repulsion mechanisms; do not abbreviate this as a completed RH proof.

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>arakelov-rh-descent</u> LEAN	Route B: a descent architecture using the stated Kim-Sarnak spectral gap, Selberg/Bost-Connes language, (S_4) , and GRH/RH implications.	A formalization project for a spectral and Arakelov descent route toward RH, with theorem-level status to be assessed from a specific manuscript and buildable release.
<u>riemann-arakelov-positivity</u> LEAN	Route A: Arakelov positivity on $(X_0(143))$, genus (13) , $(\omega^2=48/13)$, and a finite (S_4) arithmetic gate.	A Lean project formalizing an Arakelov-positivity architecture related to RH. Its individual positivity and implication statements should be cited exactly.
<u>arakelov-positivity-rh-core</u> LEAN	Central Lean core for the author's RH program, described publicly as B158 with eighteen sub-atoms and a stated classical-trio axiom footprint.	The common Lean core of the author's RH research architecture, containing named formal sub-results and declared foundational dependencies.
<u>birch-swinnerton-dyer-143</u> LEAN	BSD-related formalization for 143a1, including class number $(h(\mathbb{Q}(\sqrt{-143}))=10)$ and a claimed rank/ (L) -order relation.	A Lean development concerning explicitly named BSD data for curve 143a1; present the exact formal theorem rather than a general BSD claim.
<u>rh-p5-bridge-14</u> LEAN	A finite numerical bridge built around $(143 \cdot 13 = 1859)$, $(S_{14}=14)$, $(q_5=226)$, $(C(S_4))$, and $(h=10)$.	A finite arithmetic bridge in the author's RH/BSD program, suitable as supporting evidence only after its exact statements and dependencies are specified.

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>bost-connes</u> LEAN	Bost-Connes spectral analysis for $(X_0(143))$, with a stated finite (S_4) inequality and a sixteen-brick Lean structure.	A Lean formalization of selected Bost-Connes and finite-arithmetic components used elsewhere in the program.
<u>morningstar-project</u> LEAN	“Morning Star” certification project for stated $\text{GRH}(X_0(143))$ and $\text{BSD}(J_0(143))$ work, described with 476 equations.	A machine-certification and documentation project for the author’s $(X_0(143))$ and BSD research program; equation counts are supplementary evidence, not a theorem statement.

Complexity, algebraic geometry, and mathematical physics

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>eutheos-property</u> LEAN	Formal/computational work around the witness $(1419=3 \cdot 11 \cdot 43)$, residue conditions, barriers, and a companion complexity project.	A Lean and computational study of a specified finite witness construction used in the author’s complexity research.
<u>p-vs-np</u> LEAN	Public description states a conditional result: “if SAT is not in P, then P is not equal to NP,” with a Zenodo DOI and formal “bricks.”	A formalization of a conditional complexity statement and associated mechanisms. It should not be presented as an unconditional proof that P differs from NP.
<u>hodge-abelian-boundaries</u> LEAN	Work on measured Hodge $((2,2))$ -class obstructions on CM abelian varieties, including stated numerical rank/bound comparisons for dimensions $(3,4,5)$.	A Lean formalization of specified obstruction and rank-bound computations for CM abelian varieties; not a general proof of the Hodge Conjecture.

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>yang-mills-gap</u> LEAN	Lean work on an $SU(3)$ lattice Yang-Mills mass-gap construction at a specified coupling, using stated Bessel and Gross-Witten ingredients.	A formalization of a specified lattice gauge-theory mass-gap argument. It should be distinguished from the full Clay Yang-Mills existence-and-mass-gap problem.
<u>navier-stokes</u> LEAN	Lean repository describing a global-regularity route through ESS backward uniqueness, alongside mass-gap language and a 664-lemma count.	A Lean formalization project concerning named Navier-Stokes regularity mechanisms. Any complete global-regularity claim requires a paper stating the exact PDE setting and all analytic inputs.

Program support, computation, certification, and software

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>Certifications</u> INDEX	Repository for machine-checked Lean 4 audit certificates in the Morning Star Project.	A public certificate ledger and audit-support repository for the wider formalization program.
<u>opera-sieve</u> TYPESCRIPT / LEAN	Canonical sieve for $\backslash(S(\alpha_0=299+\pi/10)\backslash)$, combining computation with Lean verification.	A computational and Lean-verification support project for a canonical sieve used in the wider program.

REPOSITORY	PUBLICLY DESCRIBED WORK	PRESENTATION SENTENCE
<u>zerobeacon</u> PYTHON	ZeroBeacon.ai: a collision-anchored commerce router for AI agents, publicly described as an MCP server with 1,052 tools.	A separate AI-agent and commerce software project demonstrating production-oriented systems work; it is not evidence for a mathematical theorem.

Formal-methods evidence and limits

The program’s methodological value lies in its attempt to make a proof obligation inspectable. Across the Lean repositories, the public record commonly refers to source builds, theorem declarations, no-sorry checks, and axiom footprints. The Beal repository is a useful model: it separates import-free cores from Mathlib-facing wrappers, preserves the conventional $\gcd$ interface, and documents one explicit real-number quotient boundary.

The appropriate scholarly interpretation is narrow: an audited Lean theorem establishes its stated proposition relative to its definitions, imported declarations, and declared axioms. It does not automatically establish that those definitions capture a famous conjecture in its standard full form, or that a network of conditional bridges supplies every unformalized mathematical implication.

Evidence package to accompany any manuscript.

- a frozen repository release or commit hash;
- the exact Lean and Mathlib versions;
- one-command build instructions;
- a list of the principal theorem names;
- the imported theorems, assumptions, and axiom footprint;
- a conventional mathematical proof narrative and literature review;
- a statement of what remains open.

Editorial abstract for a specific submission

The following abstract is suitable for a manuscript that reports the *formal-methods contribution* of the program, rather than claiming that every associated open problem has been resolved.

We present a public Lean 4 / Mathlib formalization program for selected structures in arithmetic geometry, analytic number theory, and related mathematical physics. The work is organized as explicit, reproducible theorem developments with versioned source, build instructions, and dependency audits. Its methodology separates import-free foundational cores, public interfaces, standard imported results, finite computational certificates, and remaining formal assumptions. We illustrate the approach through formal developments involving primitive exponential Diophantine equations, modular-curve arithmetic at $(X_0(143))$, spectral and Arakelov constructions, selected BSD data, and auxiliary computational structures. The contribution is a transparent formal research architecture: each theorem is to be read with its exact hypotheses, definitions, and axiom footprint, rather than as a blanket claim of resolution of a named open problem.

Statement of suitability for arXiv sponsorship

The public record supports submitting an individual, well-scoped paper in a relevant category when that paper is topical, mathematically readable, and linked to a reproducible artifact. The author's public Lean work provides evidence of sustained engagement with formal mathematics and research software. A sponsor is not asked to certify the correctness of broad RH, BSD, Clay, or physics claims; the sponsor is asked to assess whether the particular submission is a serious, category-appropriate scholarly contribution.

arXiv's public guidance says that first submissions or new-category submissions may require endorsement, that endorsers should support authors they expect will

submit high-quality research, and that submissions should be topical and refereable. This packet is designed to document the public record behind that request while leaving the paper itself responsible for its mathematical claim.

Complete repository URL index

This index is included so that the PDF remains a usable evidence document even when a reader does not have the HTML source. Every URL below is an active hyperlink in the original public inventory.

- <https://github.com/DavidFox998/beal-conjecture>
- <https://github.com/DavidFox998/brothers-desert-proof>
- <https://github.com/DavidFox998/zerobeacon>
- <https://github.com/DavidFox998/poincare-spectral>
- <https://github.com/DavidFox998/lindelof-hypothesis-143>
- <https://github.com/DavidFox998/birch-swinnerton-dyer-143a1>
- <https://github.com/DavidFox998/eutheos-property>
- <https://github.com/DavidFox998/rh-growth-contradiction>
- <https://github.com/DavidFox998/arakelov-rh-descent>
- <https://github.com/DavidFox998/riemann-arakelov-positivity>
- <https://github.com/DavidFox998/arakelov-positivity-rh-core>
- <https://github.com/DavidFox998/birch-swinnerton-dyer-143>
- <https://github.com/DavidFox998/rh-p5-bridge-14>
- <https://github.com/DavidFox998/p-vs-np>
- <https://github.com/DavidFox998/bost-connes>
- <https://github.com/DavidFox998/yang-mills-gap>
- <https://github.com/DavidFox998/hodge-abelian-boundaries>
- <https://github.com/DavidFox998/Certifications>
- <https://github.com/DavidFox998/navier-stokes>
- <https://github.com/DavidFox998/opera-sieve>
- <https://github.com/DavidFox998/morningstar-project>

Sources and method

- [GitHub public repository API inventory](#), consulted August 20, 2026.
- [DavidFox998 public GitHub profile](#).
- Public repository metadata and README material linked in the inventory above.
- [arXiv endorsement guidance](#).
- [arXiv submission guidance](#).
- [arXiv identity and affiliation policy](#).

Scope note: this is a public-evidence document, not a peer review, theorem audit, legal certification, or bibliometric assessment. It reports public descriptions faithfully while recommending that every mathematical conclusion be evaluated from its exact statement and reproducible source.

Prepared as a public evidence packet for David J. Fox / DavidFox998. The document is intentionally claim-specific: the mathematical force of a project lies in its exact theorem and proof obligations, not in the grandeur of its title.